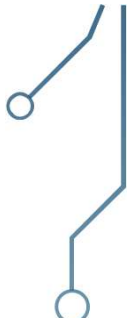


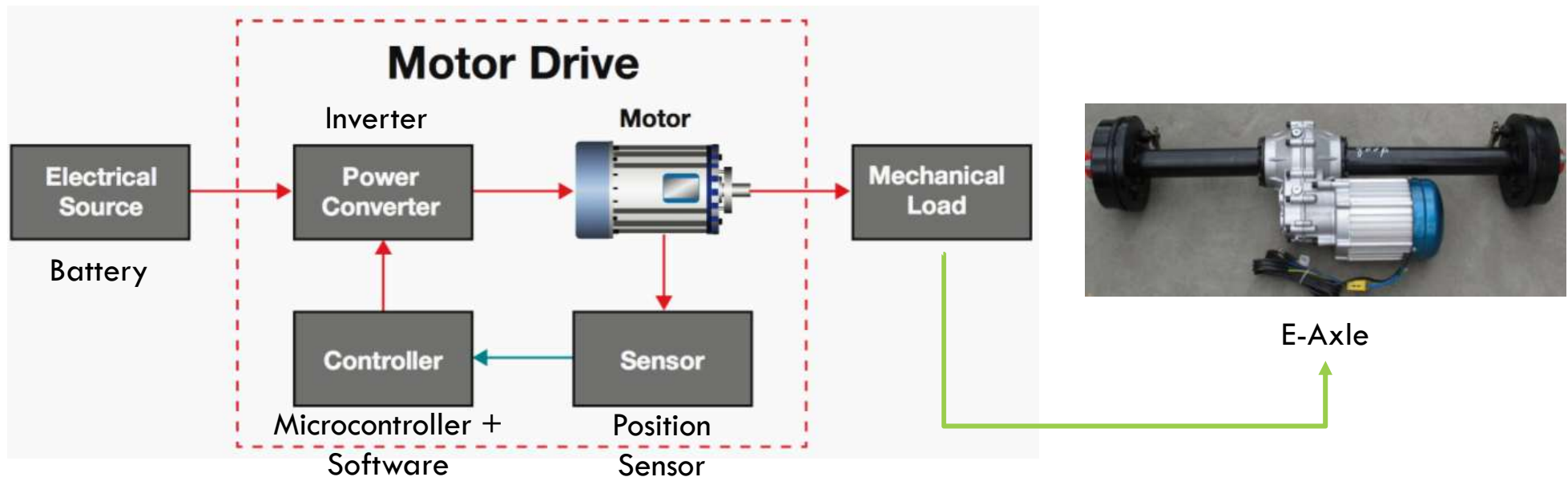


Inverter Design

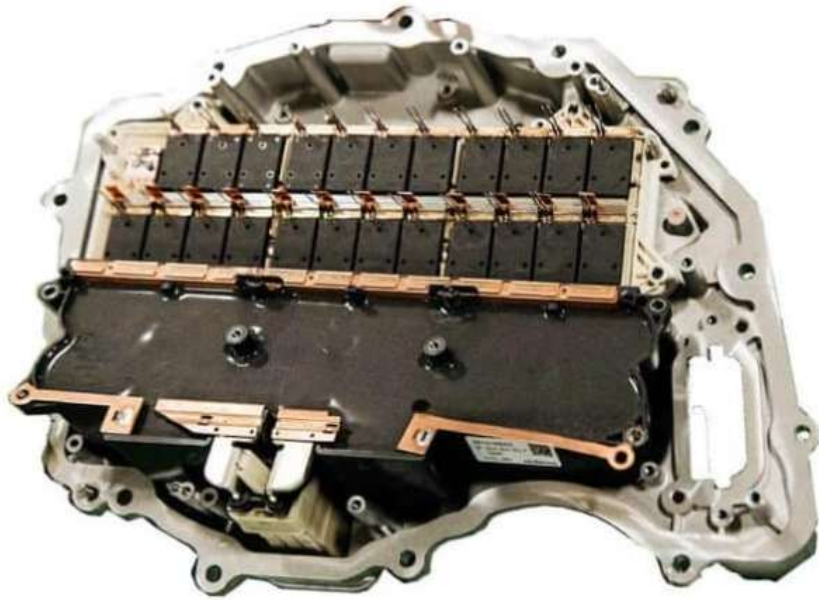
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- Basics of Electric Motors
 - Inverters
 - Simulation
- 
- 

ELECTRIC MOTORS - BASICS

An electric motor is a device **that converts electrical energy to mechanical energy**. It also can be viewed as a device that transfers energy from an electrical source to a mechanical load. The system in which the motor is located and makes it spin is called the drive, also referred to as the electric drive or motor drive.



ELECTRIC MOTORS



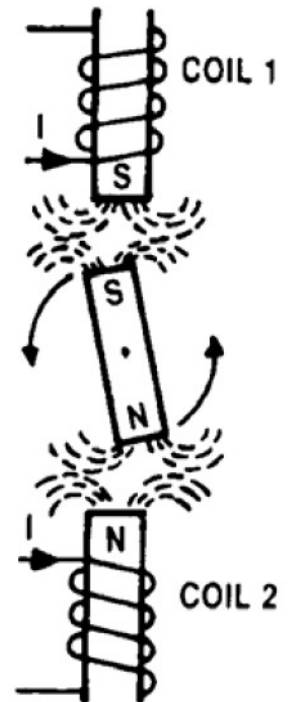
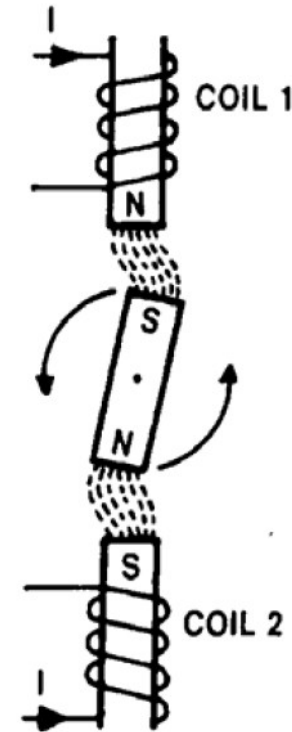
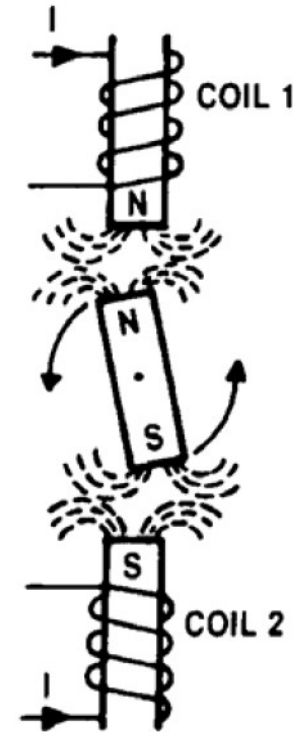
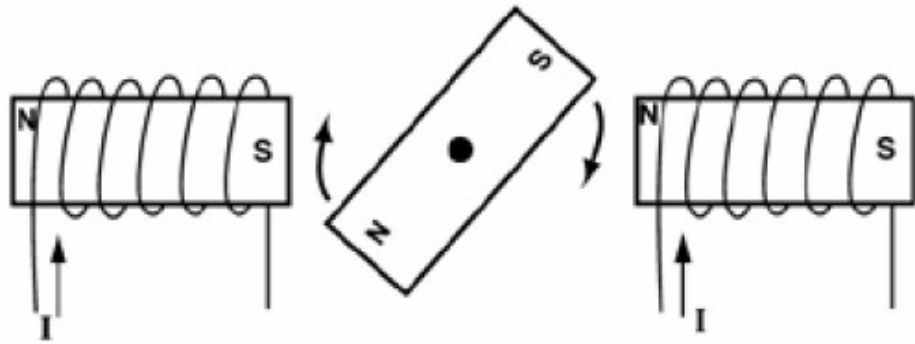
Tesla Model 3 Inverter



Tesla Model 3 Electric Motor

ELECTRIC MOTORS - PMSM

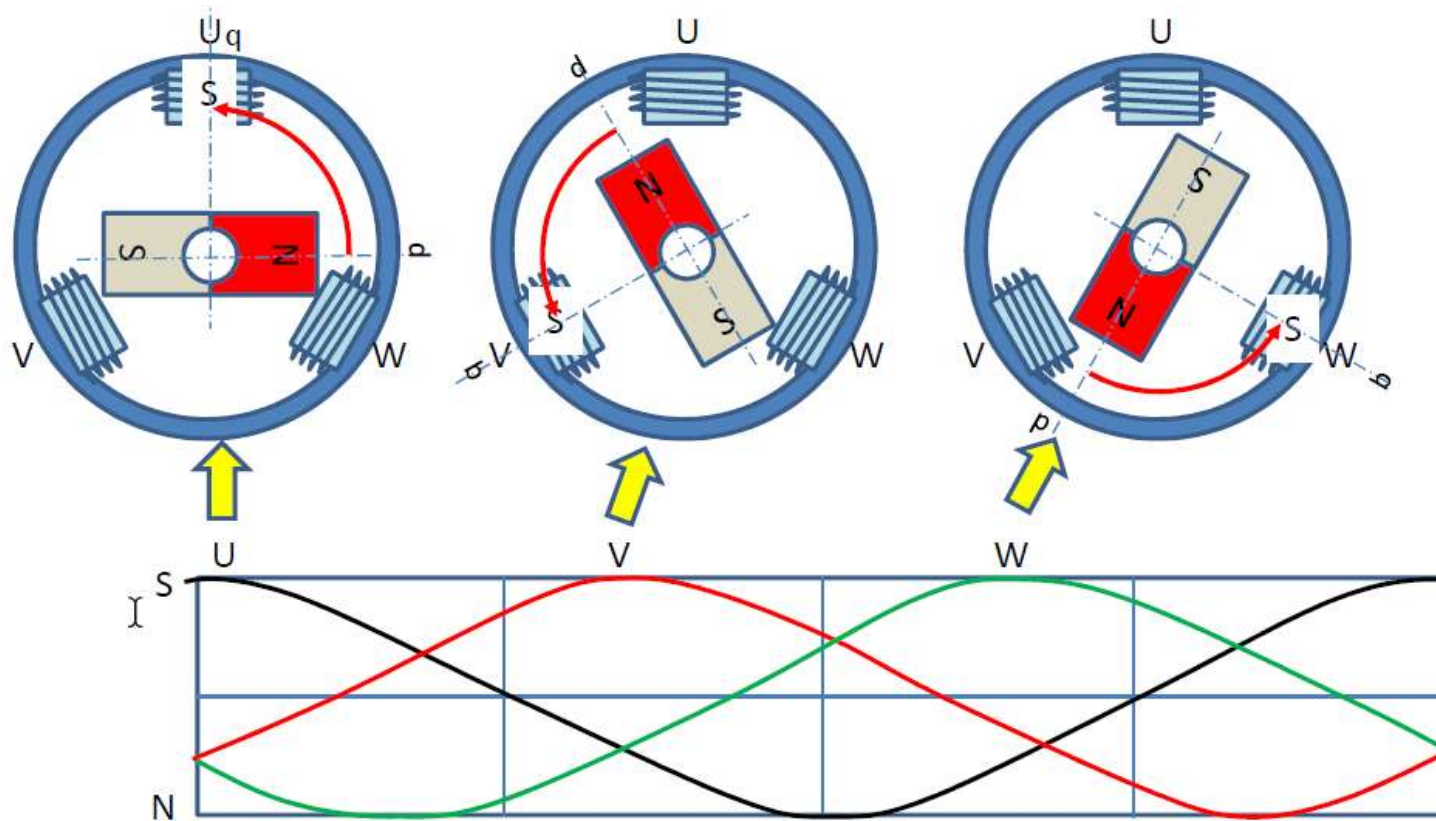
Three Phase System



- Like Poles Repel
- Opposite Poles Attract

ELECTRIC MOTORS - PMSM

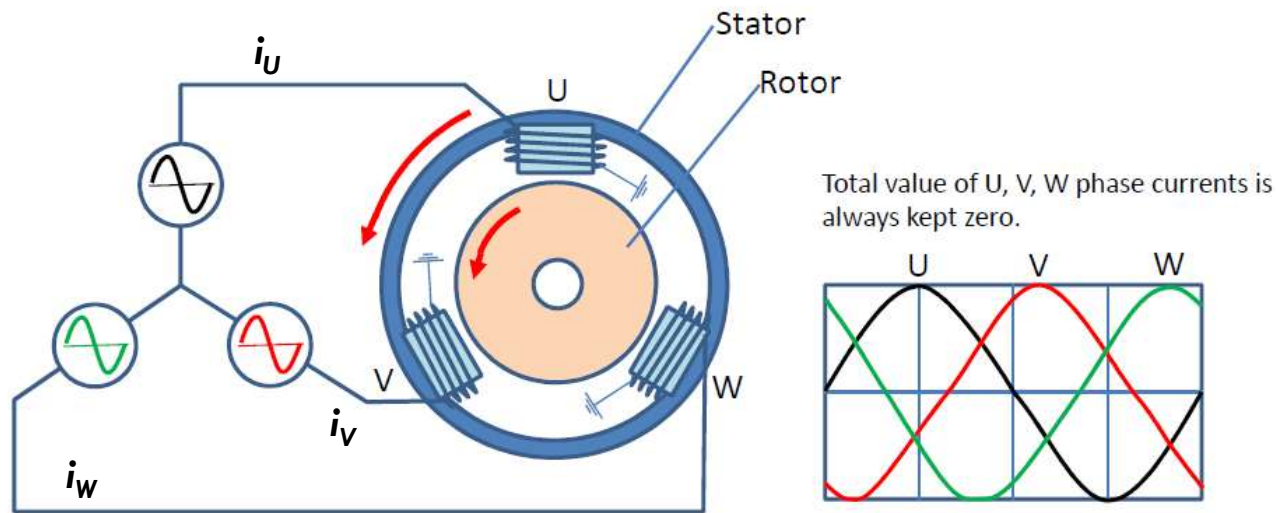
Three Phase System



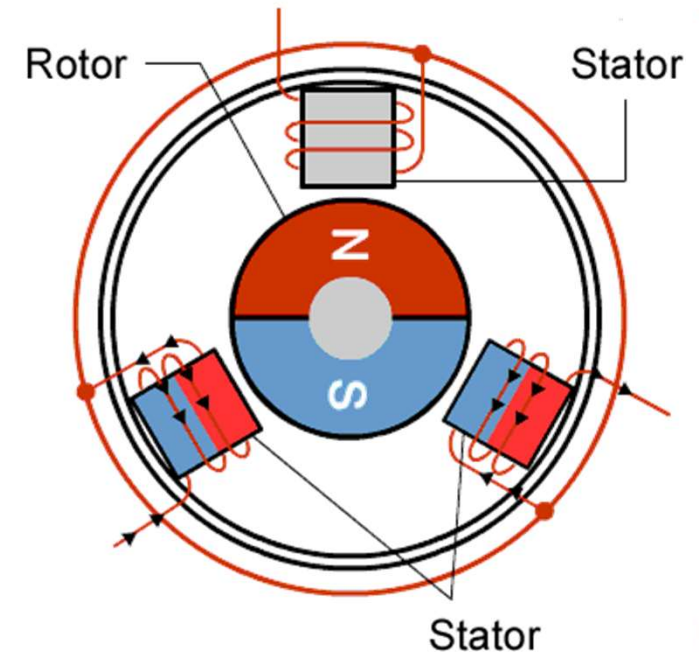
We must measure the speed and the position of the rotor to control the motion of the motor accurately!

ELECTRIC MOTORS - PMSM

Three Phase System



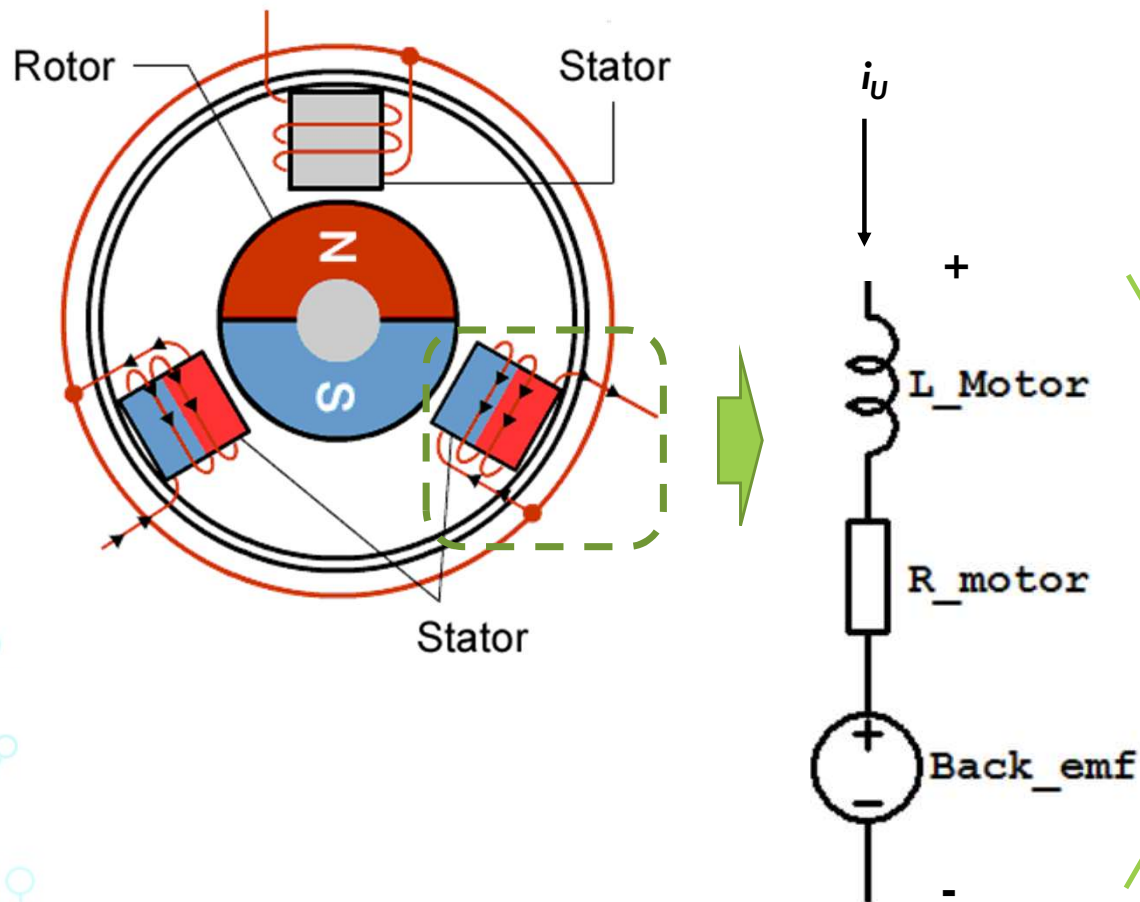
120° Phase Difference between each Phase



A current is injected into each winding (copper wire) of the electric motor with a 120° phase shift. This current generates an electromagnetic field/force (EMF) which repels or pulls the rotor depending on the rotor position and the stator current.

ELECTRIC MOTORS - PMSM

Three Phase System



Each motor Winding can be modeled with an inductance, resistance and a voltage source modeling the Back-EMF of the motor. The back emf voltage is proportional to the speed of the motor: the faster motor spins, the larger back-emf voltage gets

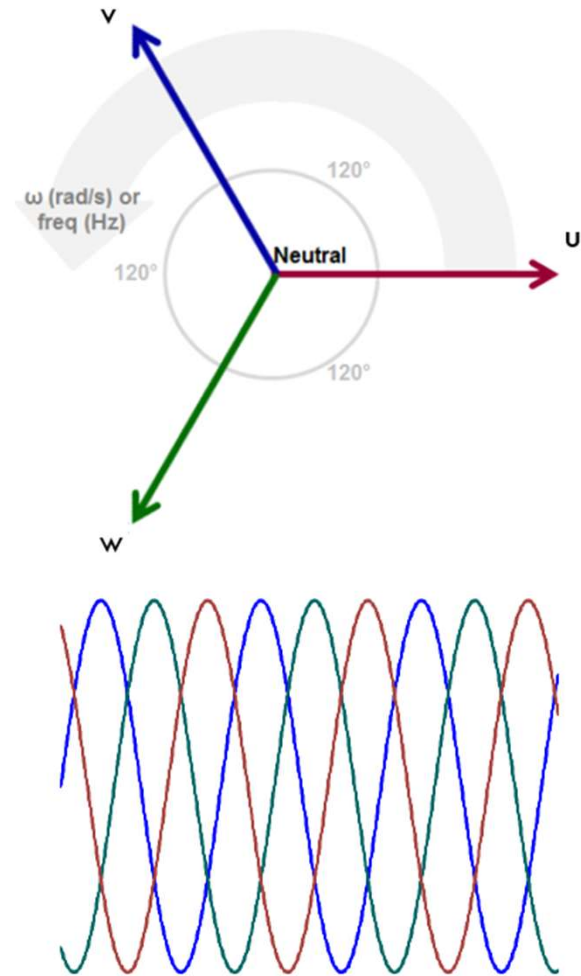
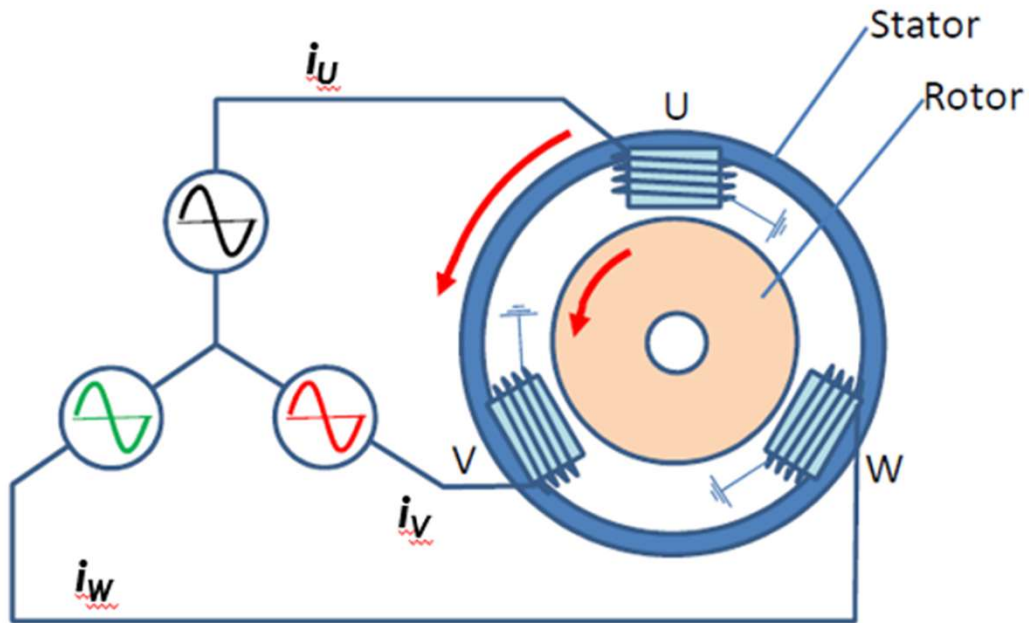
$$V = i_U \cdot R_motor + i_U \cdot Z_{L_motor} + K_e \omega_r$$

K_e : Back EMF constant

ω_{er} : rotor speed

ELECTRIC MOTORS - PMSM

Three Phase System



When electrically observed as three different voltage-vs-time waveforms, the “rotating” voltage vector system appears as three sinusoidal waveforms. The waveforms are at the same fixed frequency but out of phase with respect to each other by 120°

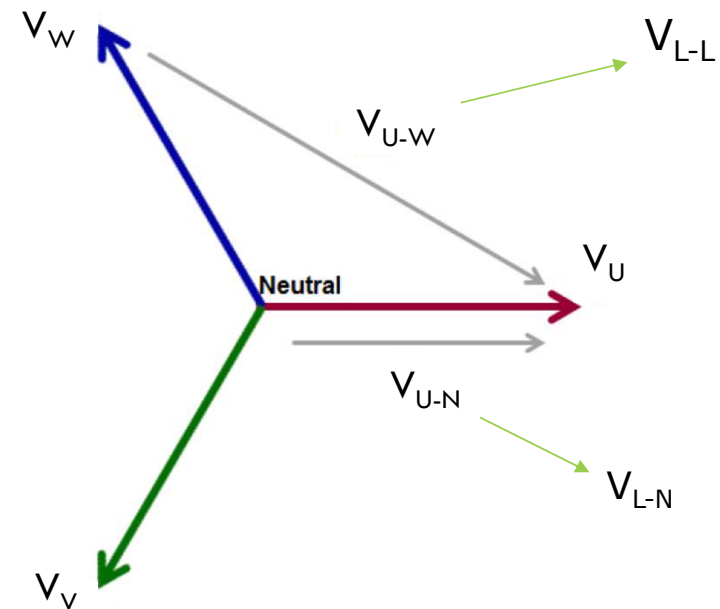
ELECTRIC MOTORS - PMSM

Three Phase System

In a three-phase ac system, voltage can be measured either from line-to-line (phase-to-phase) or from line-to-neutral (phase-to-neutral).

Note that line-to-line may also be referred to as phase-to-phase. The convention for indicating a line-to-line measurement is to show the voltage symbol "V" followed by a subscripted "line-line" designation, with the first line shown as the "to" line and the second line shown as the "from" line. Thus, measuring from line U to line V is expressed as V_{U-V} or simply V_{AB} . This is different from the vector notation for UV, which defines a vector direction from V to U.

Measurements of line-to-line **RMS voltage** (not peak) magnitudes may be converted to line-to-neutral RMS voltage magnitudes using the simple formula, $V_{L-N} = V_{L-L} / \sqrt{3} \rightarrow 120^\circ$ between two line voltages!



ELECTRIC MOTORS - PMSM

Three Phase System

In a three-phase system, we express the rated ac voltage in V_{rms} , but the rated voltage value is a line-to-line voltage value. In a single-phase system, the rated voltage value is a line-to-neutral value because there may not be a second line to reference a voltage to in a single-phase system. Therefore, if a three-phase system is rated at 480 V or 480 Vac, this means that the three-phase system is 480 V_{rms} *line-to-line*. Applying that knowledge allows us to calculate other voltage values, as follows:

$$V_{pk-L} = \sqrt{2} * V_{ac}, \text{ or } \sqrt{2} * V_{rms} = 679 \text{ V}$$

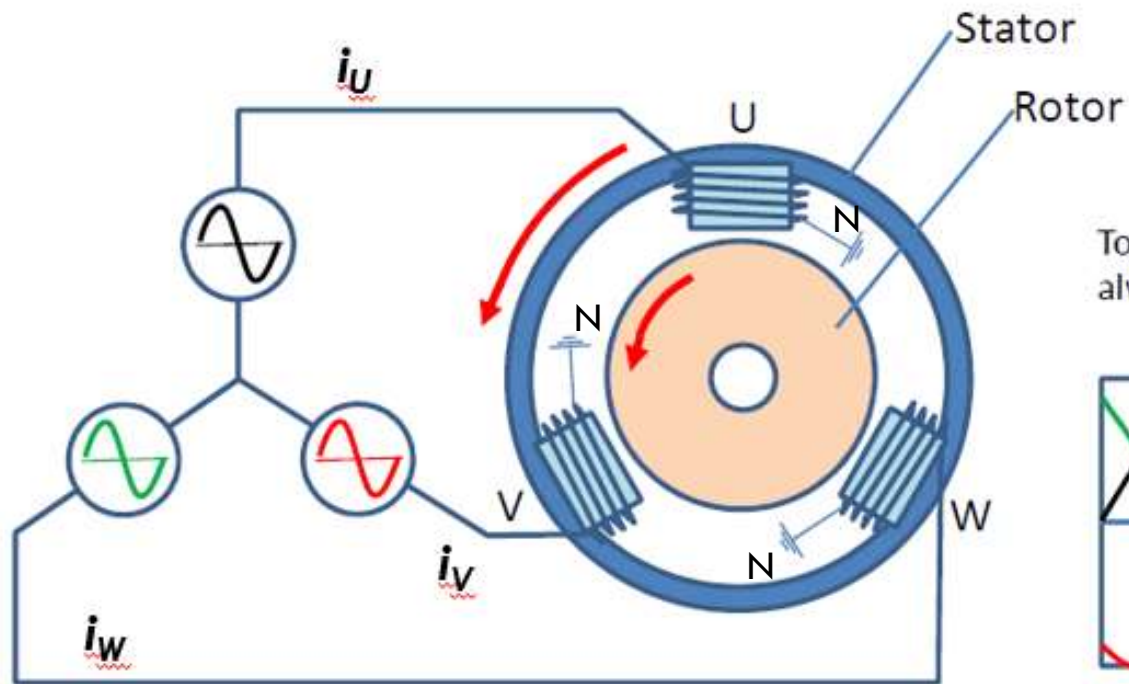
$$V_{pk-pk-L} = 2 * V_{pk-L} = 1358 \text{ V}$$

ELECTRIC MOTORS - PMSM

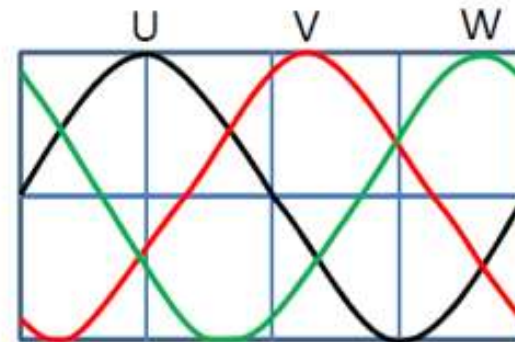
Three Phase System

For a balanced system ($i_N = 0$):

$$i_U + i_V + i_W = 0$$



Total value of U, V, W phase currents is always kept zero.



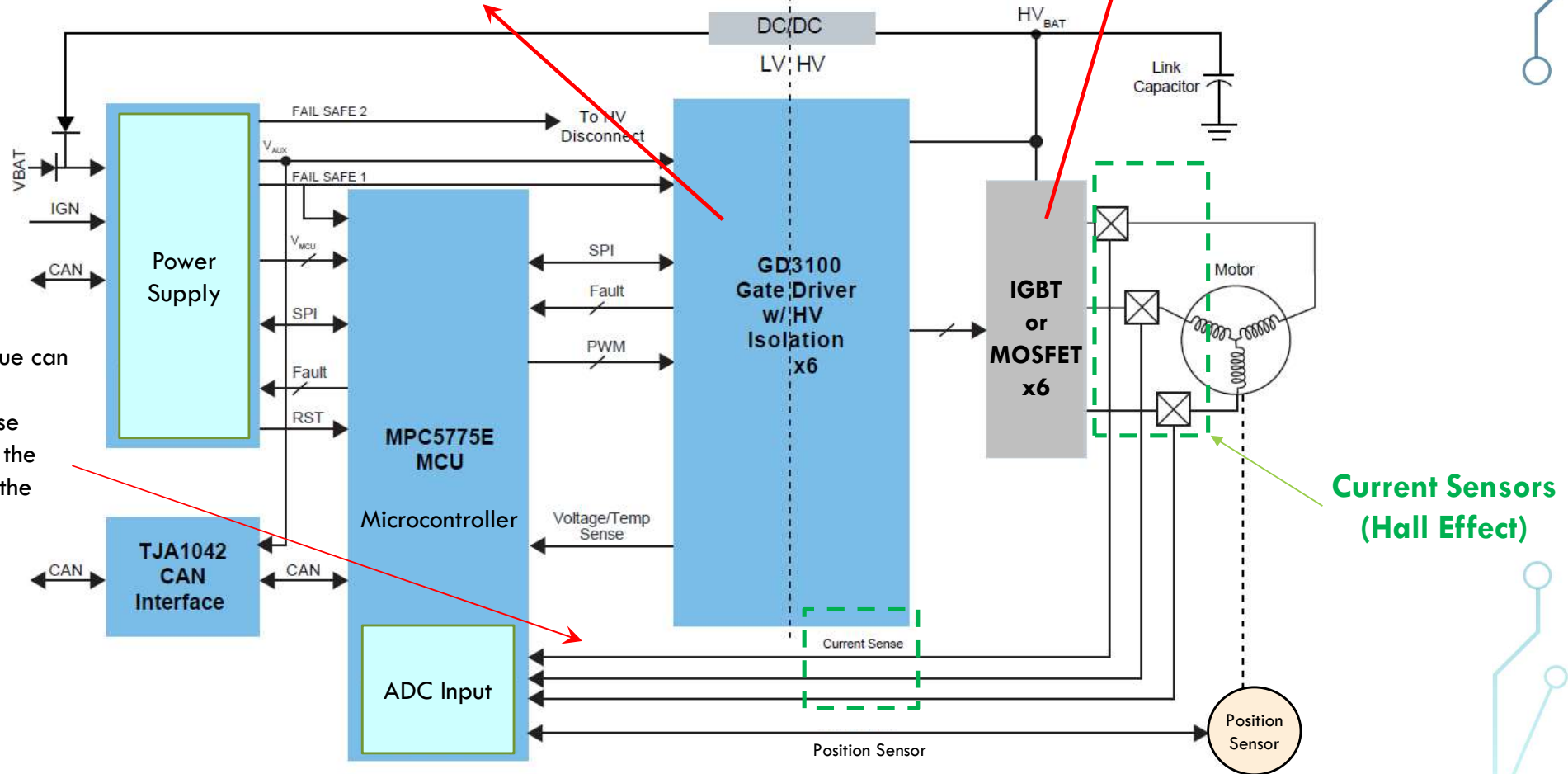
How are the phase currents generated? → Inverters

THREE PHASE INVERTERS

Torque Control

This block has already been investigated!

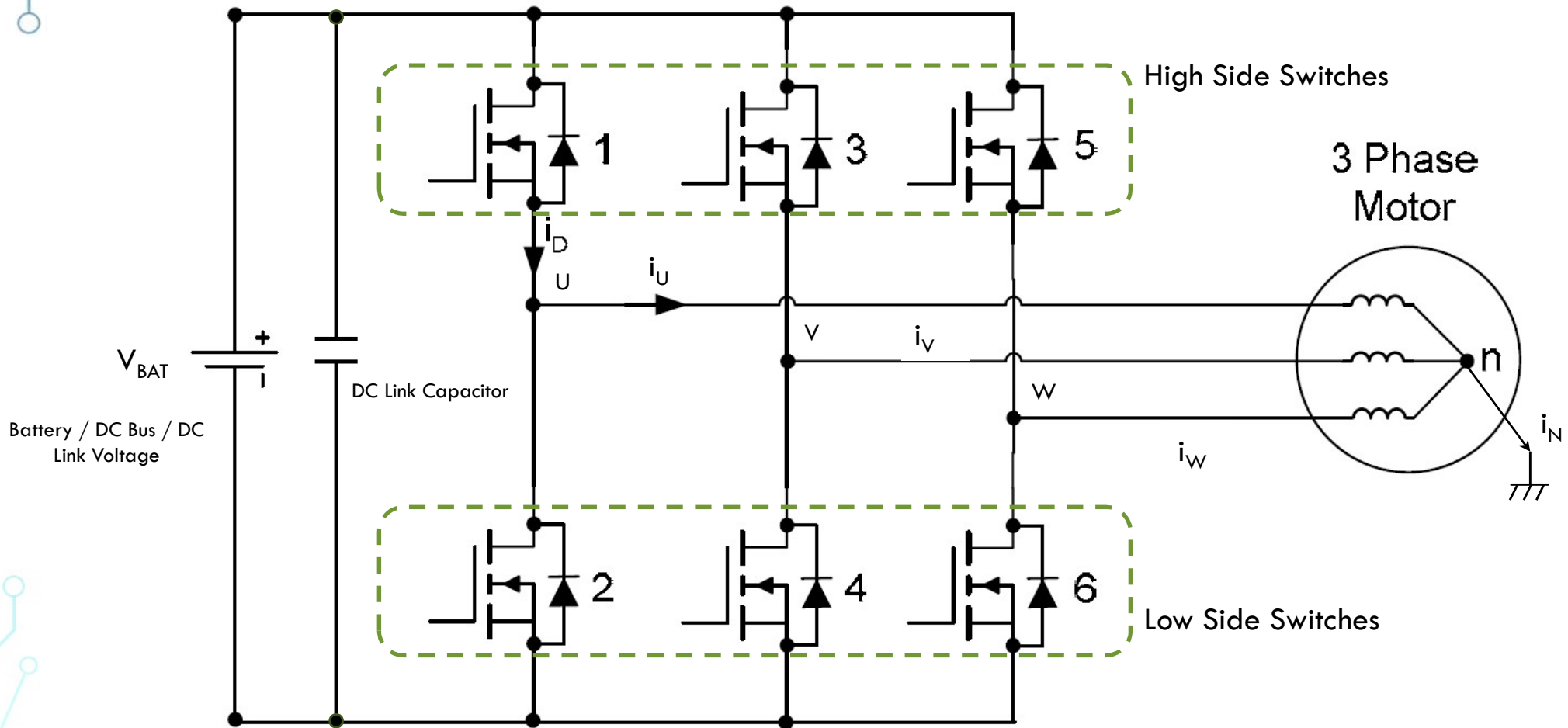
This block will be shown at the next slide!



Electric Motor Torque can be calculated by measuring the phase current. The larger the current, the larger the torque.

THREE PHASE INVERTERS

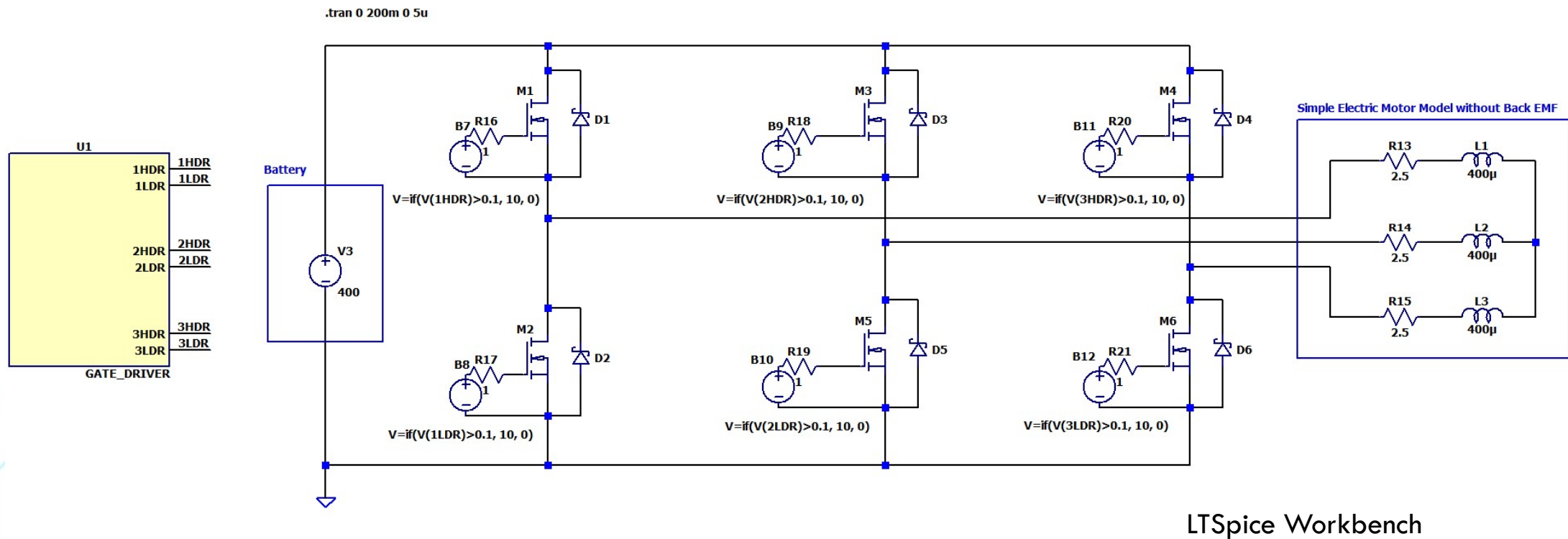
Two-Level Three Phase Inverter



THREE PHASE INVERTERS

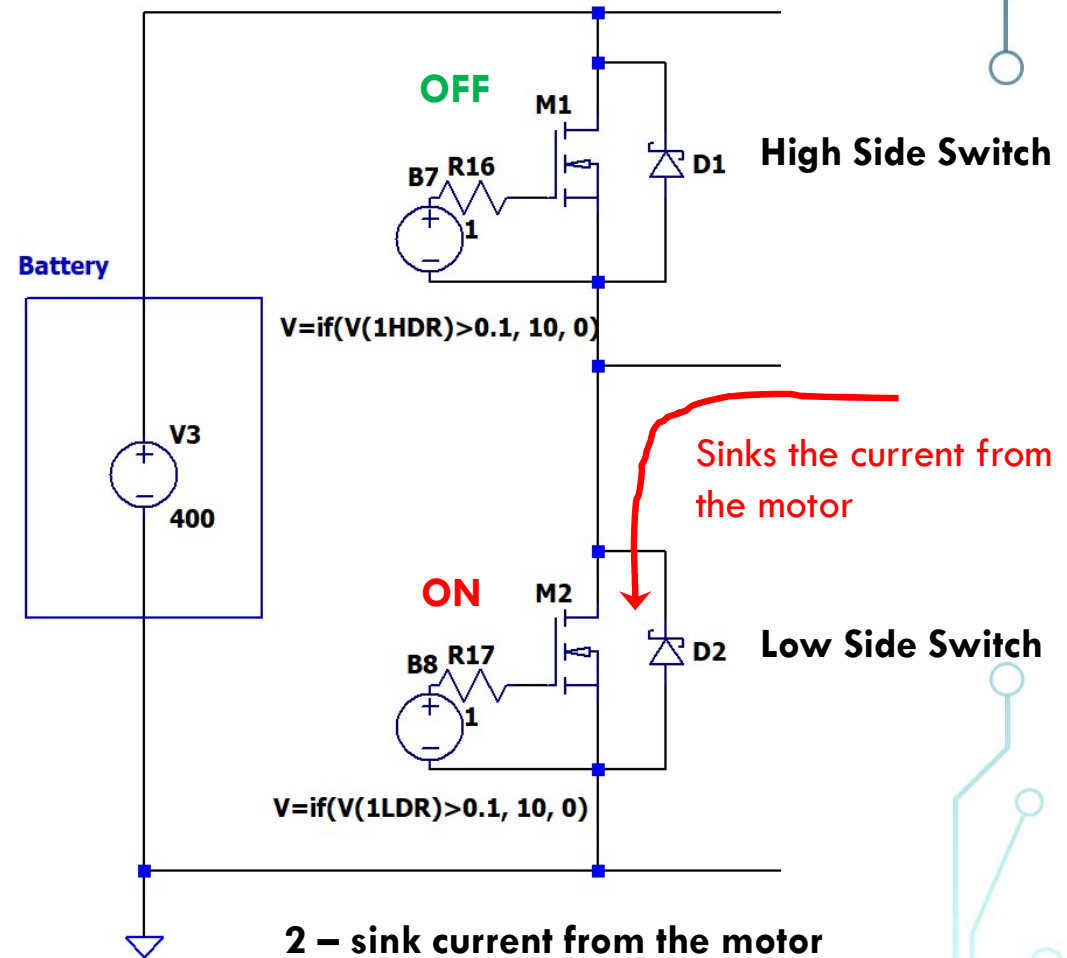
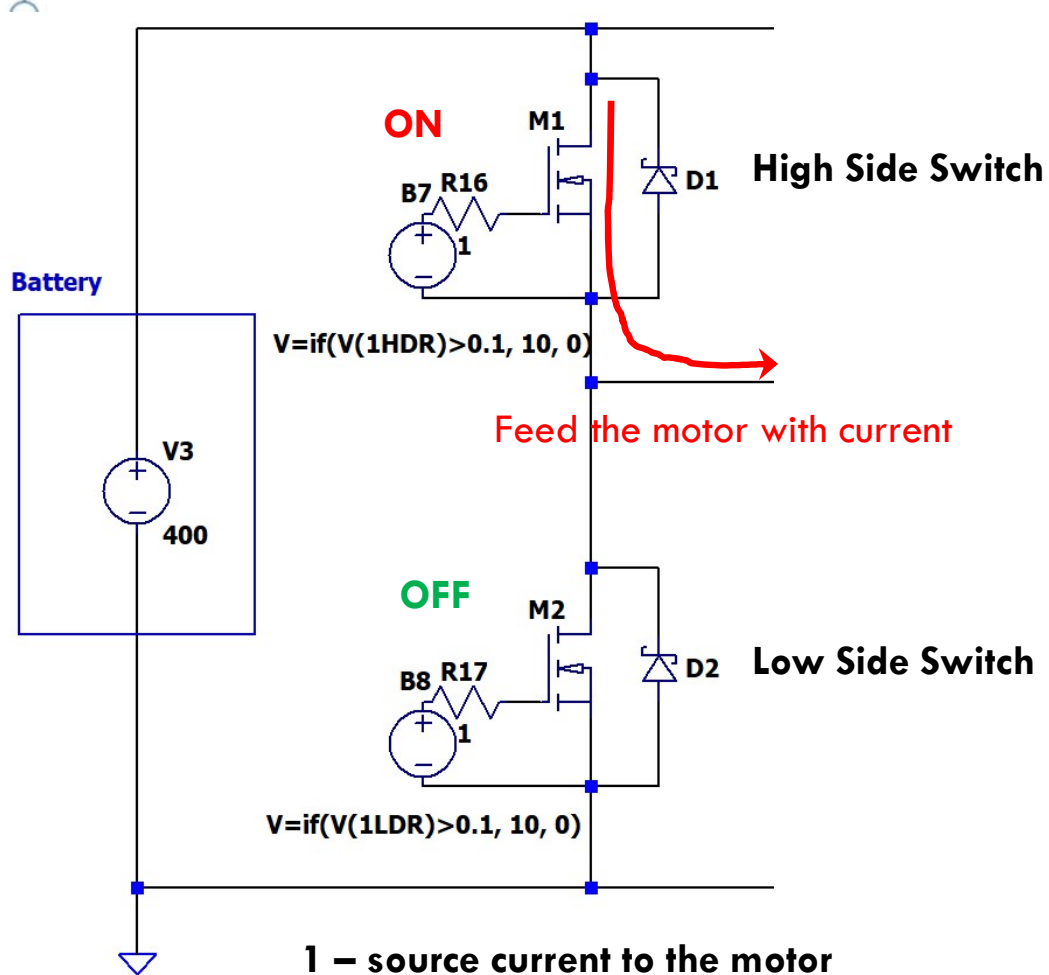
Three Phase Inverter – Working Principle

THREE PHASE INVERTER



THREE PHASE INVERTERS

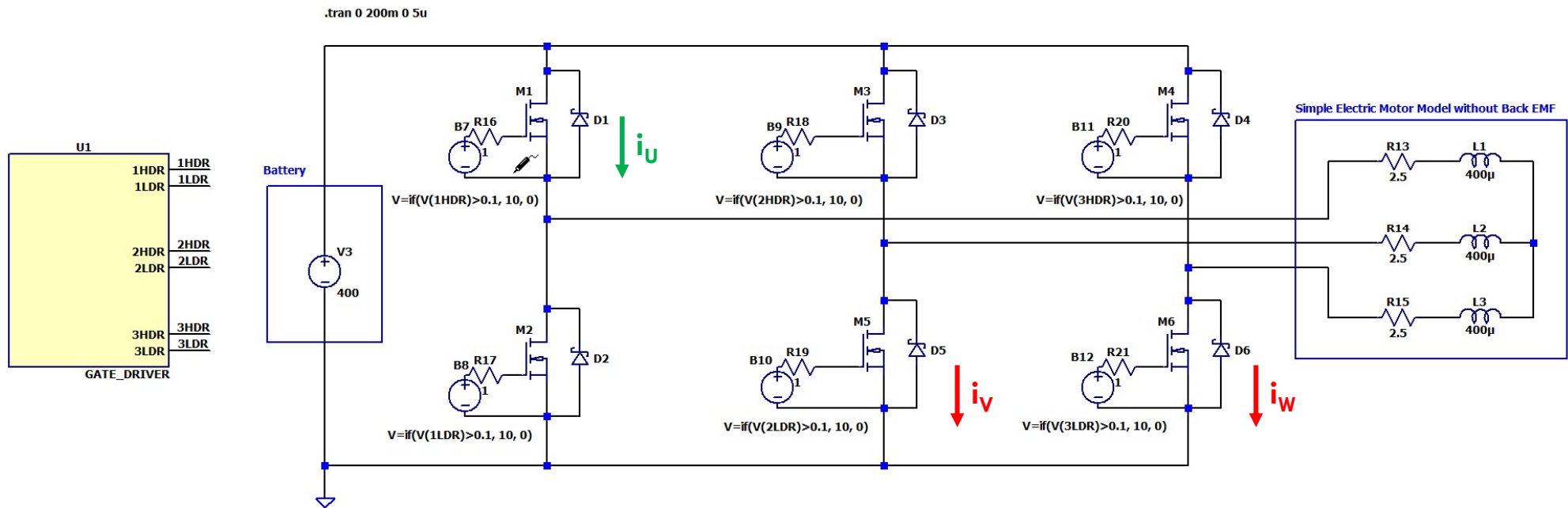
Three Phase Inverter – Complementary Switch



THREE PHASE INVERTERS

Three Phase Inverter – Example Current Flow

THREE PHASE INVERTER

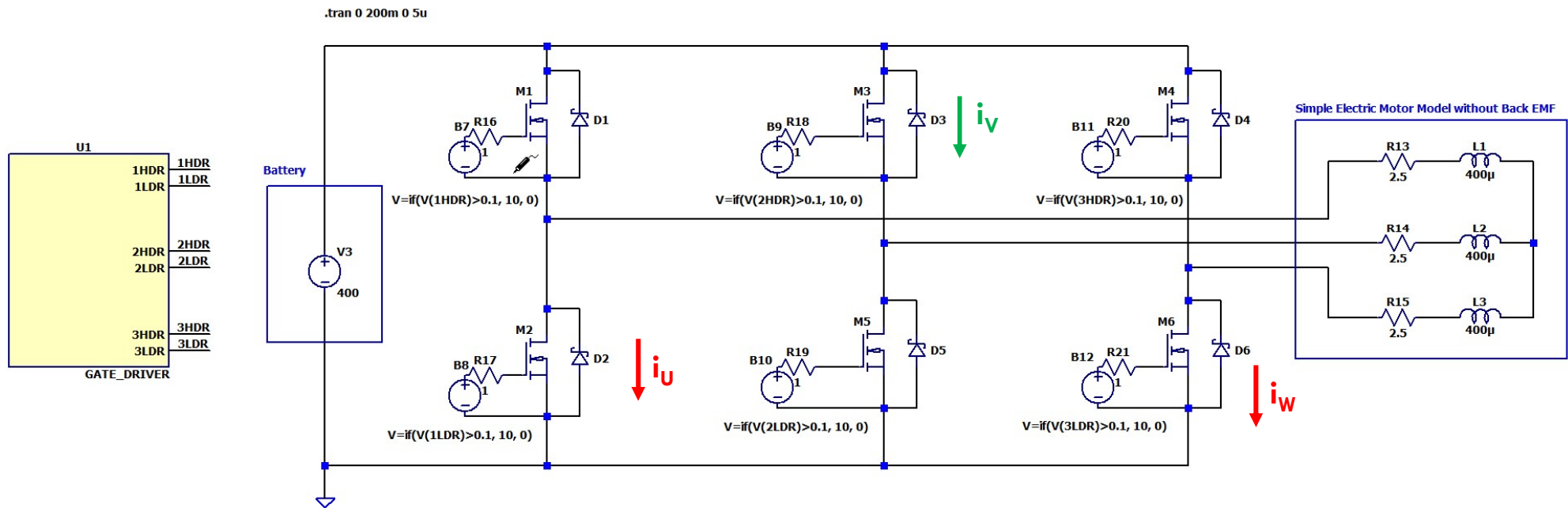


Sum of currents flowing into the motor must be equal to the sum of the currents flowing out of the motor (Kirchhoff Law)

THREE PHASE INVERTERS

Three Phase Inverter – Example Current Flow

THREE PHASE INVERTER



Sum of currents flowing into the motor must be equal to the sum of the currents flowing out of the motor (Kirchhoff Law)

e.g., $i_v = i_u + i_w$

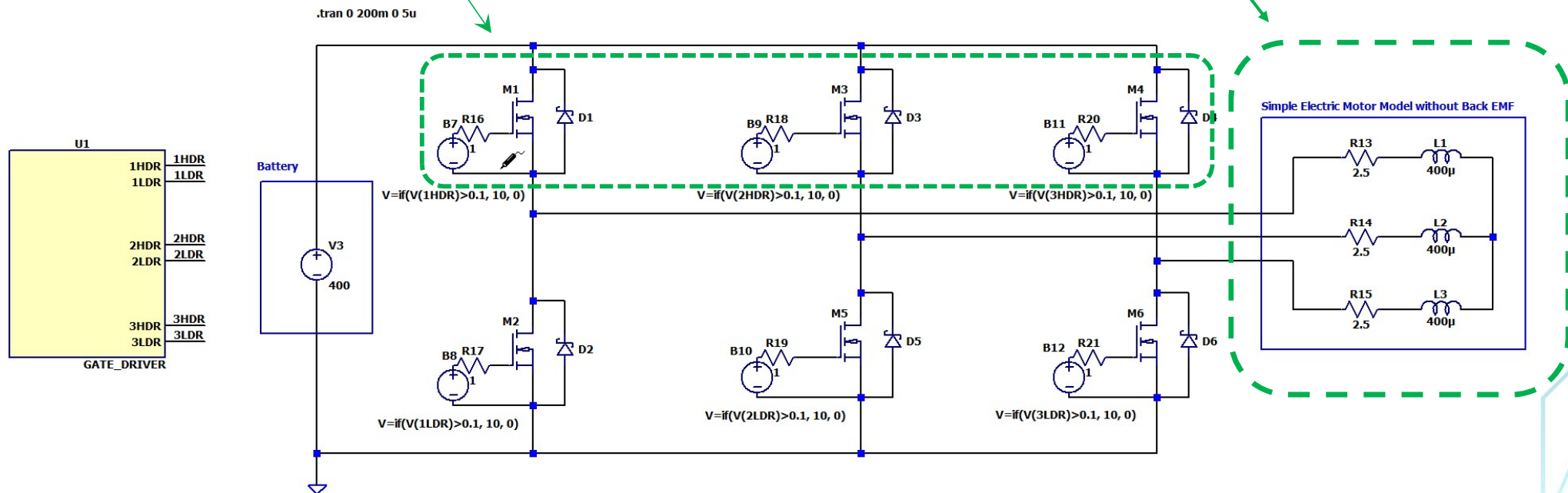
THREE PHASE INVERTERS

Three Phase Inverter – Generating a Sine Wave

Bootstrap, Charge Pump or Isolated
DCDC Converter

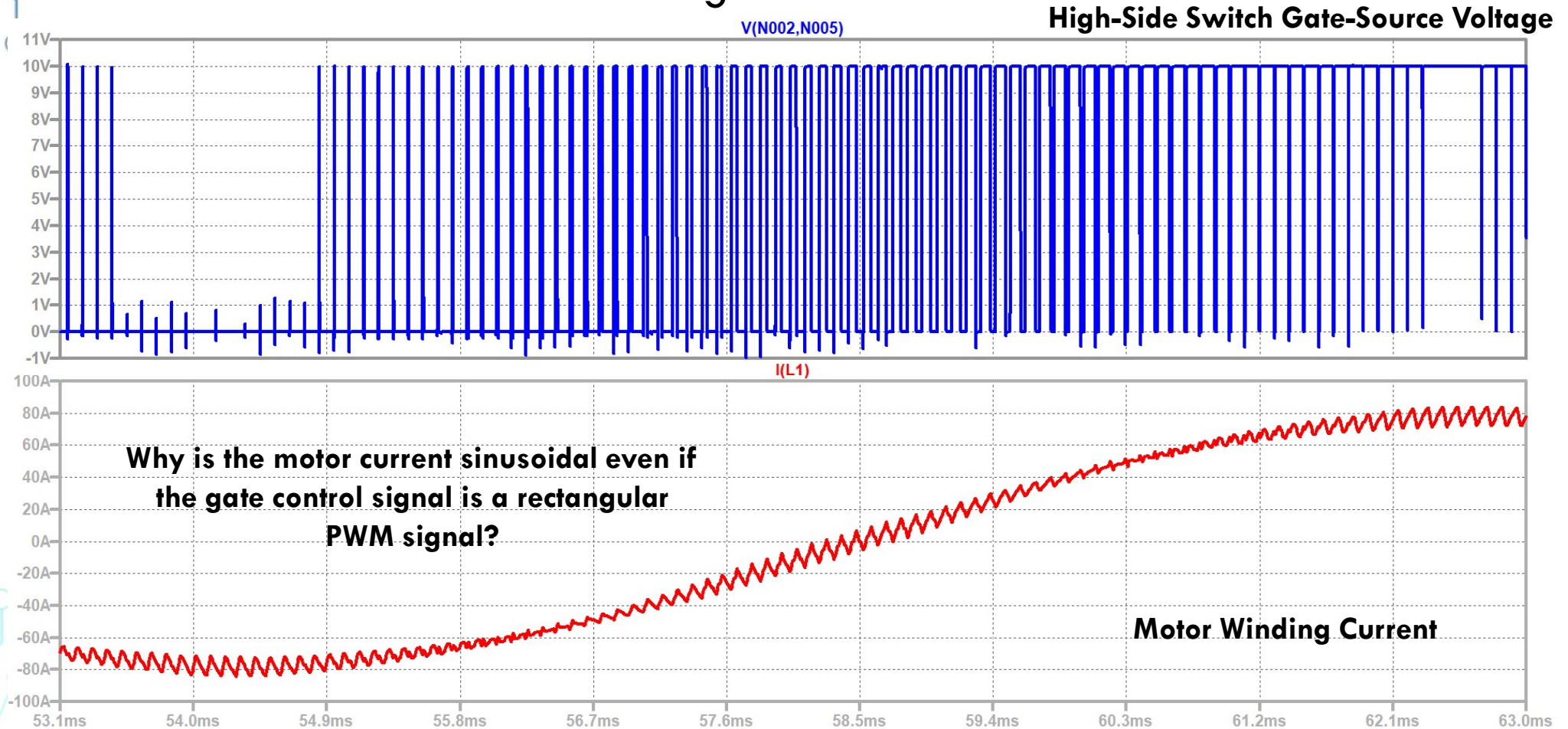
The motor acts as a low-pass filter due to its large
inductance → the PWM at the gate of the switches
becomes a sinusoidal current at the motor

THREE PHASE INVERTER



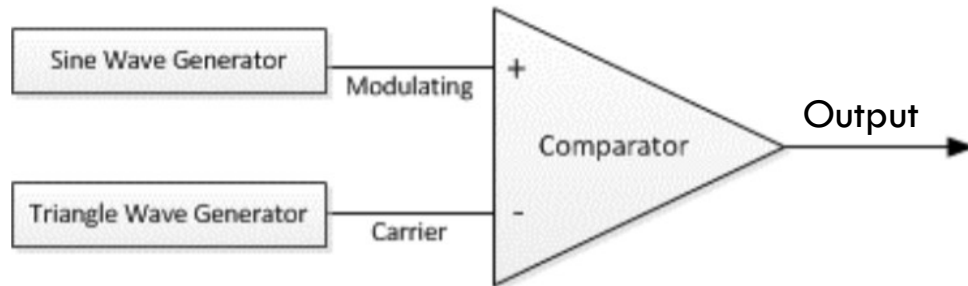
THREE PHASE INVERTERS

Three Phase Inverter – Generating a Sine Wave

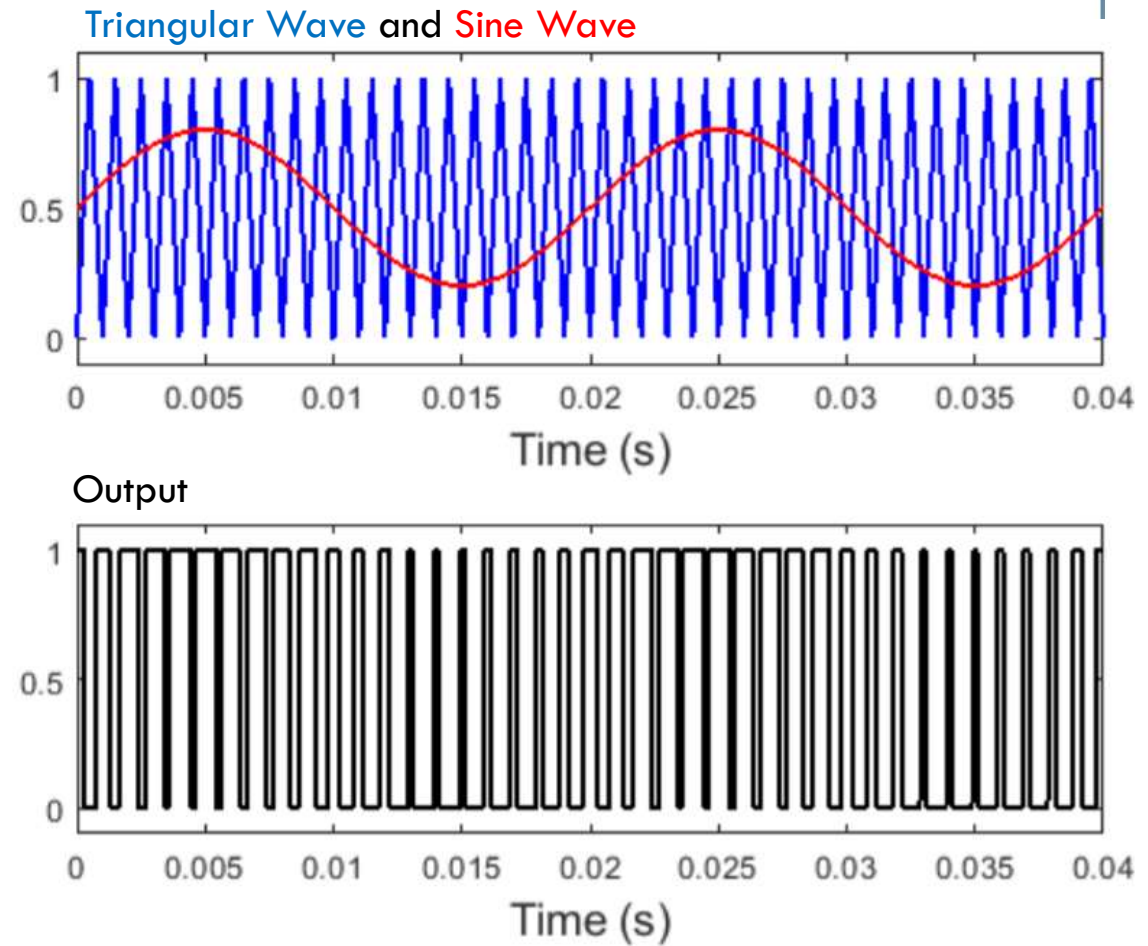


THREE PHASE INVERTERS

Three Phase Inverter – Generating a Sine Wave



Output Signal normally generated in a microcontroller



THREE PHASE INVERTERS

Three Phase Inverter

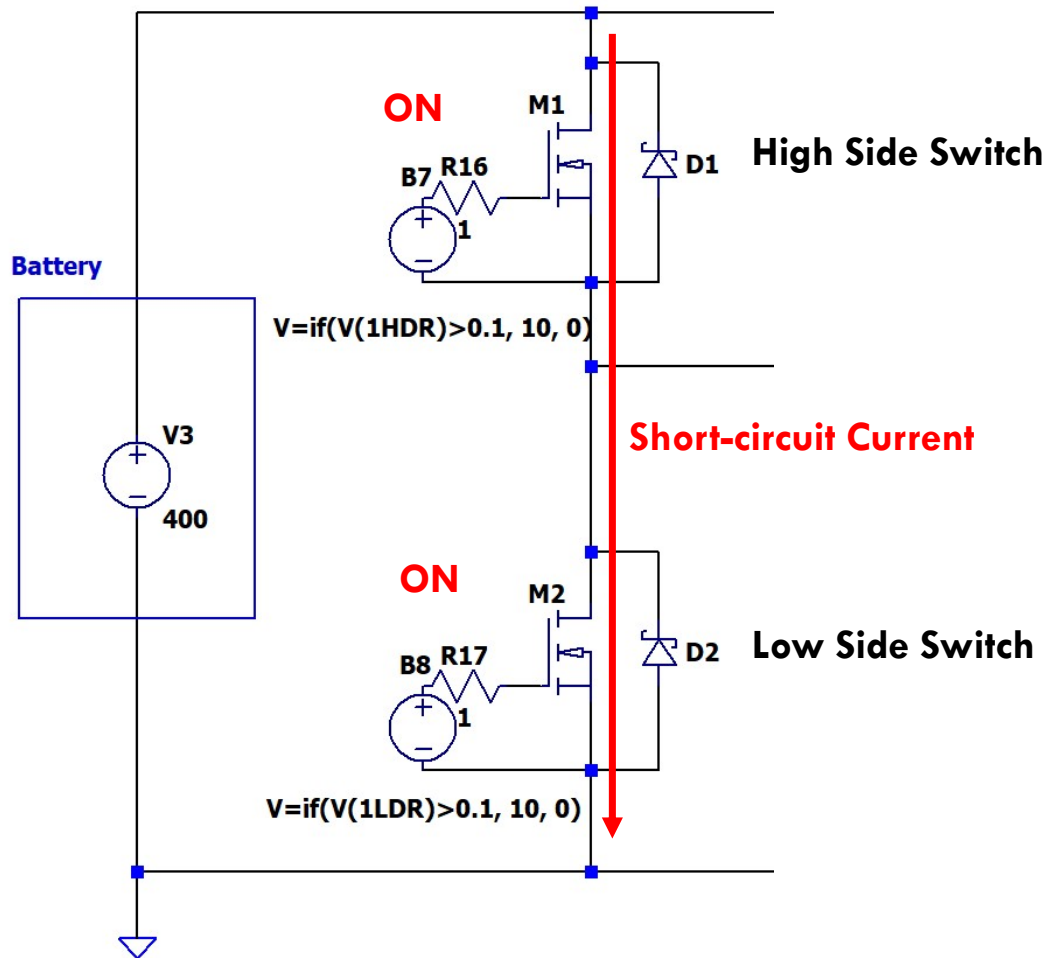
In a three-phase inverter using sinusoidal PWM (SPWM), the modulation process generates PWM gate signals that drive the gate-to-source voltage of the MOSFETs. The duty cycle of each PWM signal is proportional to the instantaneous amplitude of the sinusoidal reference waveform.

When the MOSFET gate-to-source voltage exceeds the threshold voltage V_{th} , the device turns on and conducts. Depending on which switch in the inverter leg is active, current is either supplied from the DC bus to the motor phase winding or returned from the winding back to the DC bus.

Because the motor windings are inductive, they act as a low-pass filter for the high-frequency PWM voltage. As a result, the high-frequency switching components are attenuated, and the phase currents follow a nearly sinusoidal waveform corresponding to the fundamental component of the applied voltage.

THREE PHASE INVERTERS

Three Phase Inverter – Shoot-through

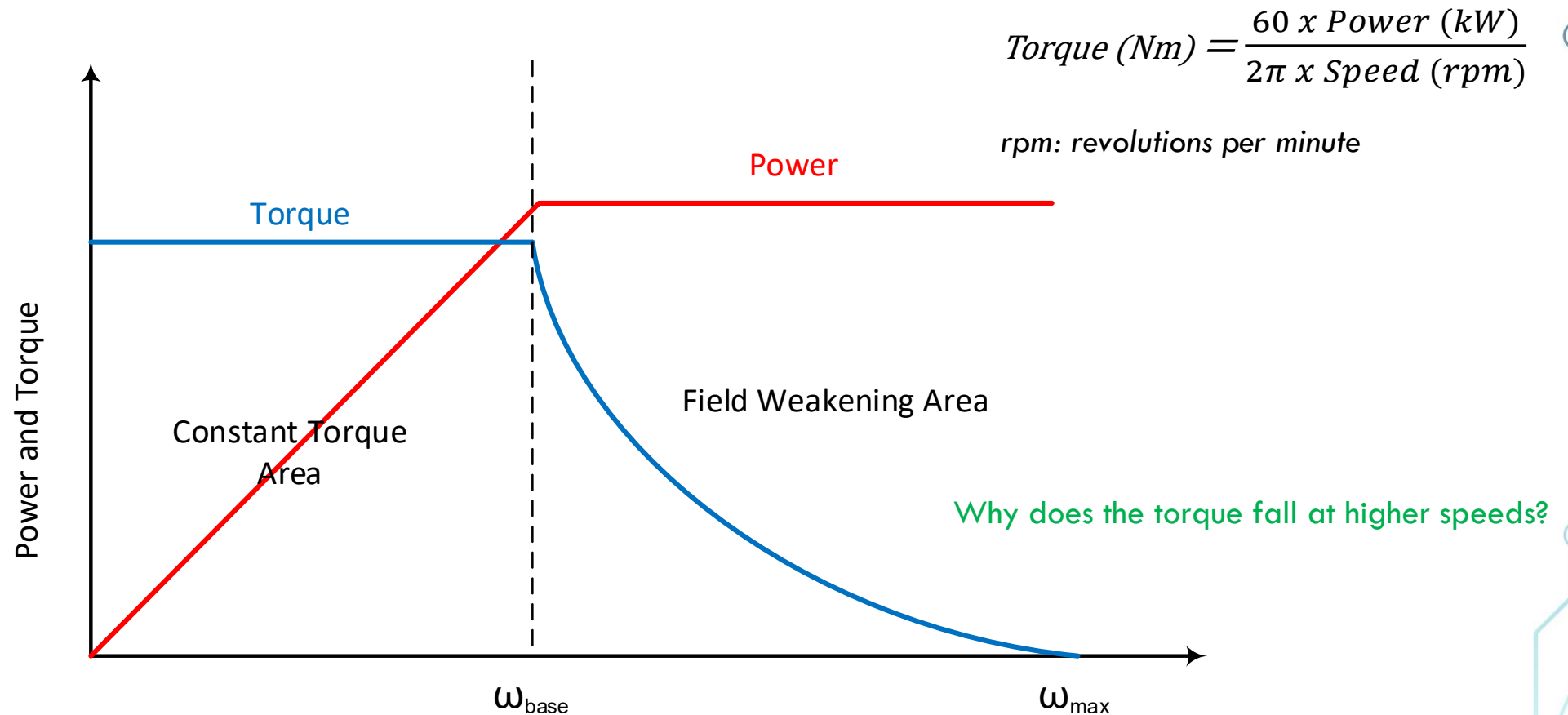


If the HS and the LS switches turned on at the same time during switching, the battery poles are short circuited → short circuit errors for some microseconds, severe increase of MOSFET power losses!!! This can be prevented by introducing a dead time during switching!

Complementary Half Bridge Stage and
Dead Time already covered in the
previous lectures: **LDOs and DCDC**
Controllers

THREE PHASE INVERTERS

Three Phase Inverter – Power and Torque Characteristics

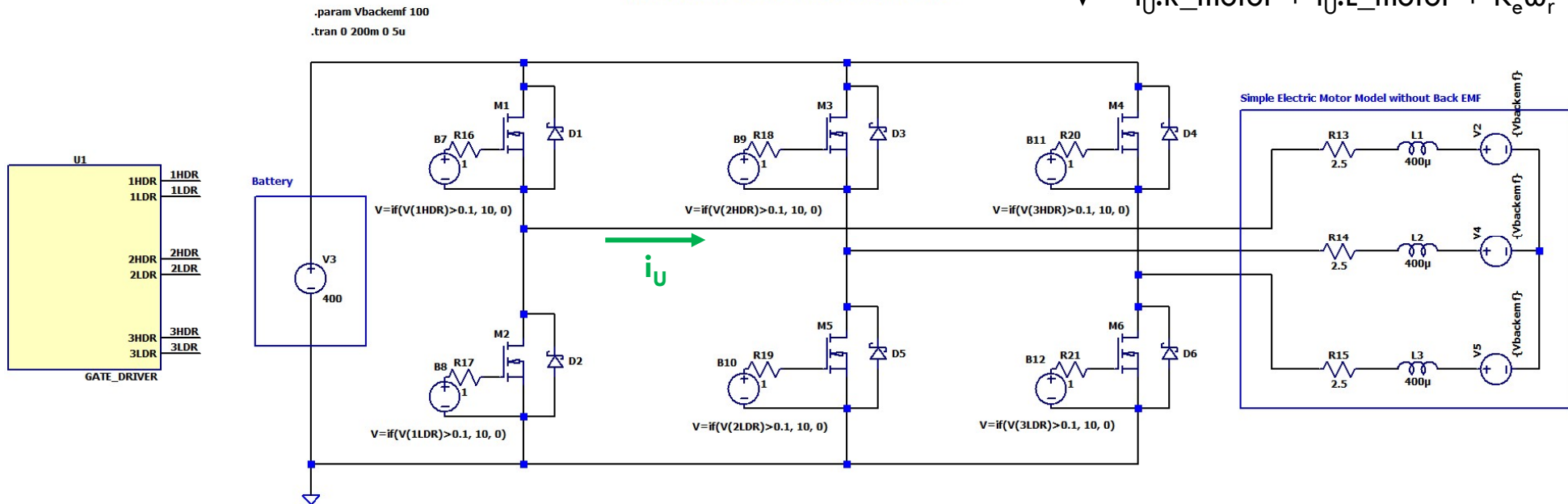


THREE PHASE INVERTERS

Three Phase Inverter – Power and Torque Characteristics

THREE PHASE INVERTER

$$V = i_U \cdot R_{\text{motor}} + i_U \cdot L_{\text{motor}} + K_e \omega_r$$



The back EMF voltage becomes larger at higher speeds which limits the current flowing into the motor (stator) winding. The torque is proportional to phase current. Therefore, this leads at the same time to a decrease of torque.

EXAMPLE

A three-phase, two-level inverter is used to drive a permanent-magnet motor under steady operating conditions. Each inverter phase leg consists of an upper MOSFET and a lower MOSFET, each including an antiparallel body diode. To avoid shoot-through, a fixed dead time is inserted between the gate signals of complementary MOSFETs.

DC Bus or DC Link Voltage $\rightarrow V_{DC} = 400 \text{ V}$

Phase Current $\rightarrow I_{ph} = 12 \text{ A}$

Switching Frequency $\rightarrow f_{sw} = 20 \text{ kHz}$

Dead Time $\rightarrow t_d = 1.5 \mu\text{s}$

MOSFET On-State Resistance $\rightarrow R_{ds,on} = 18 \text{ m}\Omega$

Turn-on and Turn-off Energy $\rightarrow E_{on} = 70 \mu\text{J}, E_{off} = 50 \mu\text{J}$

Junction-to-Case Thermal Resistance $\rightarrow R_{JC} = 0.4^\circ\text{C/W}$

Case-to-Heatsink Thermal Resistance $\rightarrow R_{CS} = 0.2^\circ\text{C/W}$

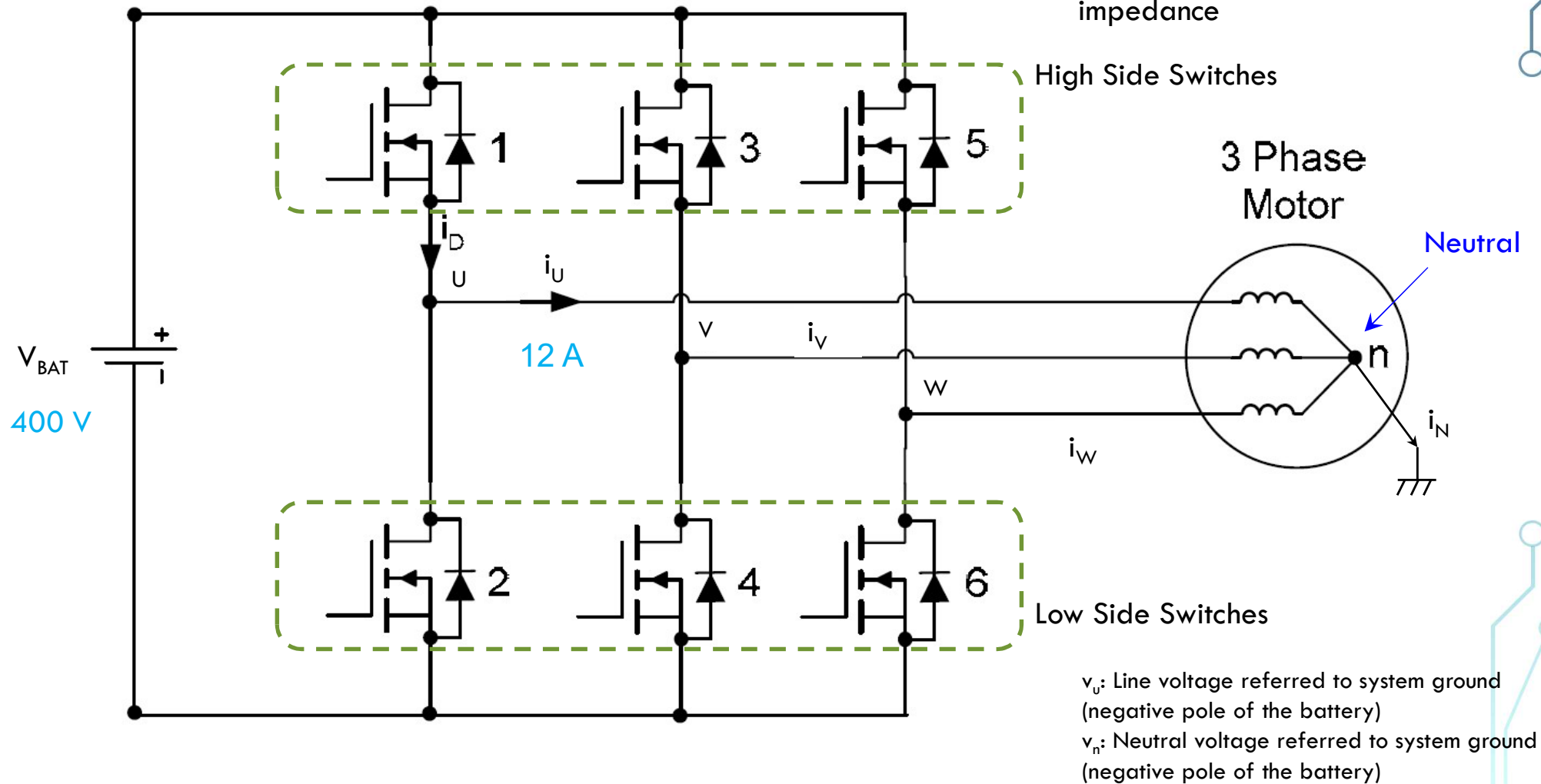
Heatsink-to-Coolant Thermal Resistance $\rightarrow R_{S,cool} = 3.5^\circ\text{C/W}$

Coolant Temperature $\rightarrow T_{cool} = 40^\circ\text{C}$

Body Diode Forward Voltage $\rightarrow V_D = 1.0 \text{ V}$

THREE PHASE INVERTERS

Two-Level Three Phase Inverter



EXAMPLE

a) Calculate the instantaneous power loss of MOSFETs and body diode

- MOSFET

$$P_{cond} = I_{ph}^2 R_{ds,on}$$

$$P_{cond} = 12^2 \cdot 0.018 = 144 \cdot 0.018 = 2.59 \text{ W}$$

- Body Diode Loss during Dead Time → both HS and LS MOSFETs turned off

$$P_{BD} = I_{ph} \cdot V_D = 12 \cdot 1.0 = 12 \text{ W}$$

Average Body Diode Loss:

$$P_{BD} = 12 \cdot \frac{2 \cdot 1.5}{50} = 0.72 \text{ W}$$

Explanation: $f_{sw} = 20 \text{ kHz} \rightarrow T_{sw} = 50 \mu\text{s}$ and two dead times per switching period!

Switching Loss per MOSFET:

$$P_{SW} = f_{SW} \cdot (E_{on} + E_{off}) = 20000 \cdot (70 + 50) \cdot 10^{-6} = 2.4 \text{ W} \rightarrow \text{please see Basics of MOSFET Power Loss}$$

b) Calculate the MOSFET junction temperature

$$R_{J,cool} = R_{JC} + R_{CS} + R_{S,cool} = 4.1^\circ\text{C/W}$$

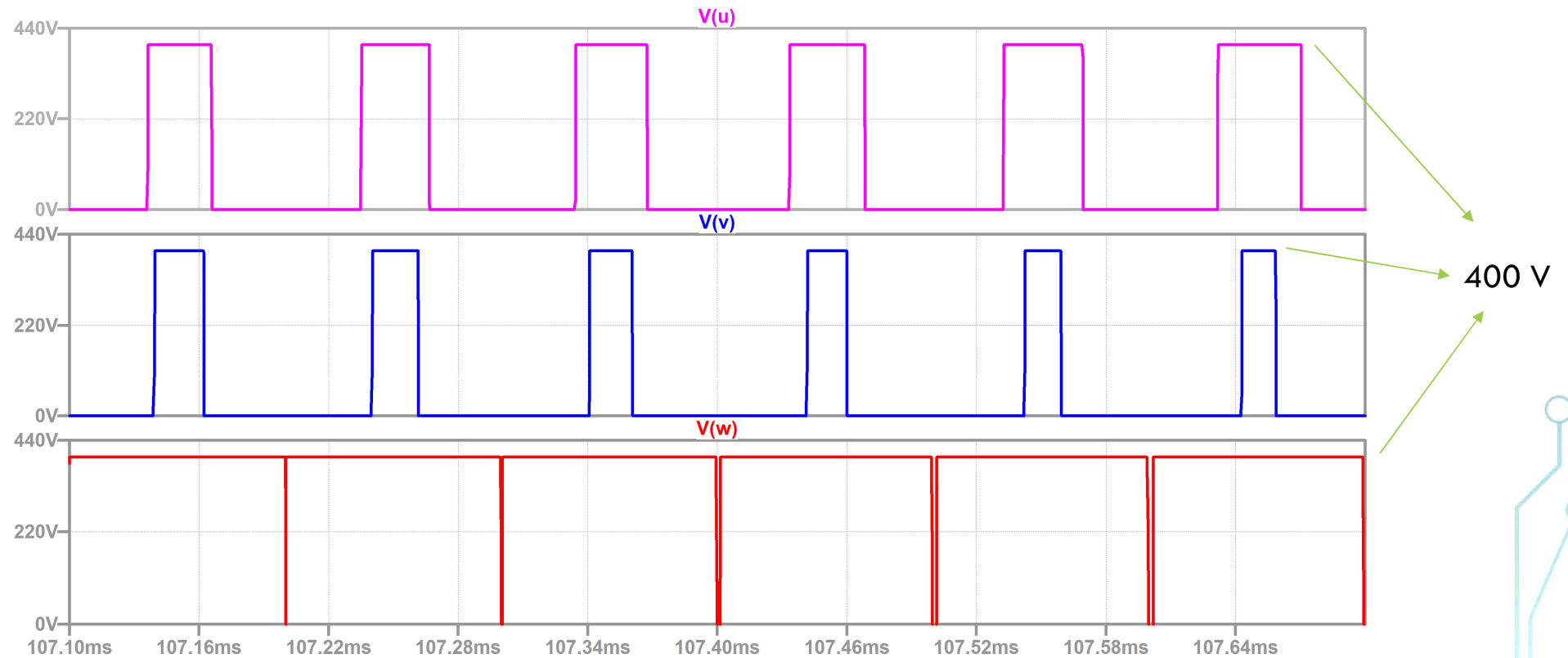
EXAMPLE

$$\Delta T_J = P_{Total} \cdot R_{J,cool} = 5.72 \cdot 4.1 \approx 23.45^\circ\text{C}$$

$$T_J = 40 + 23.45 = 63.45^\circ\text{C}$$

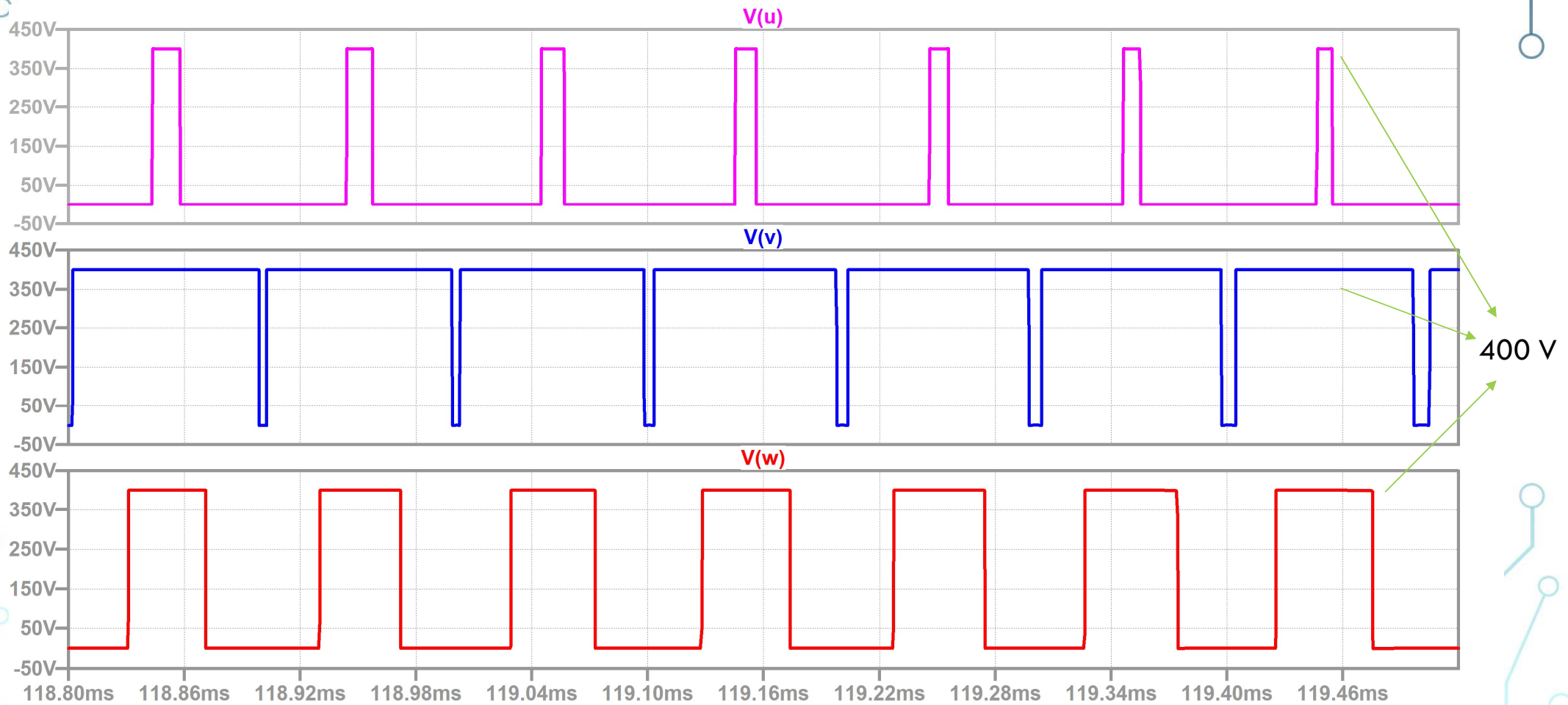
c) Plot the phase voltages and currents

Phase (Line-to-Ground) Voltages between 107.1 ms and 107.68 ms



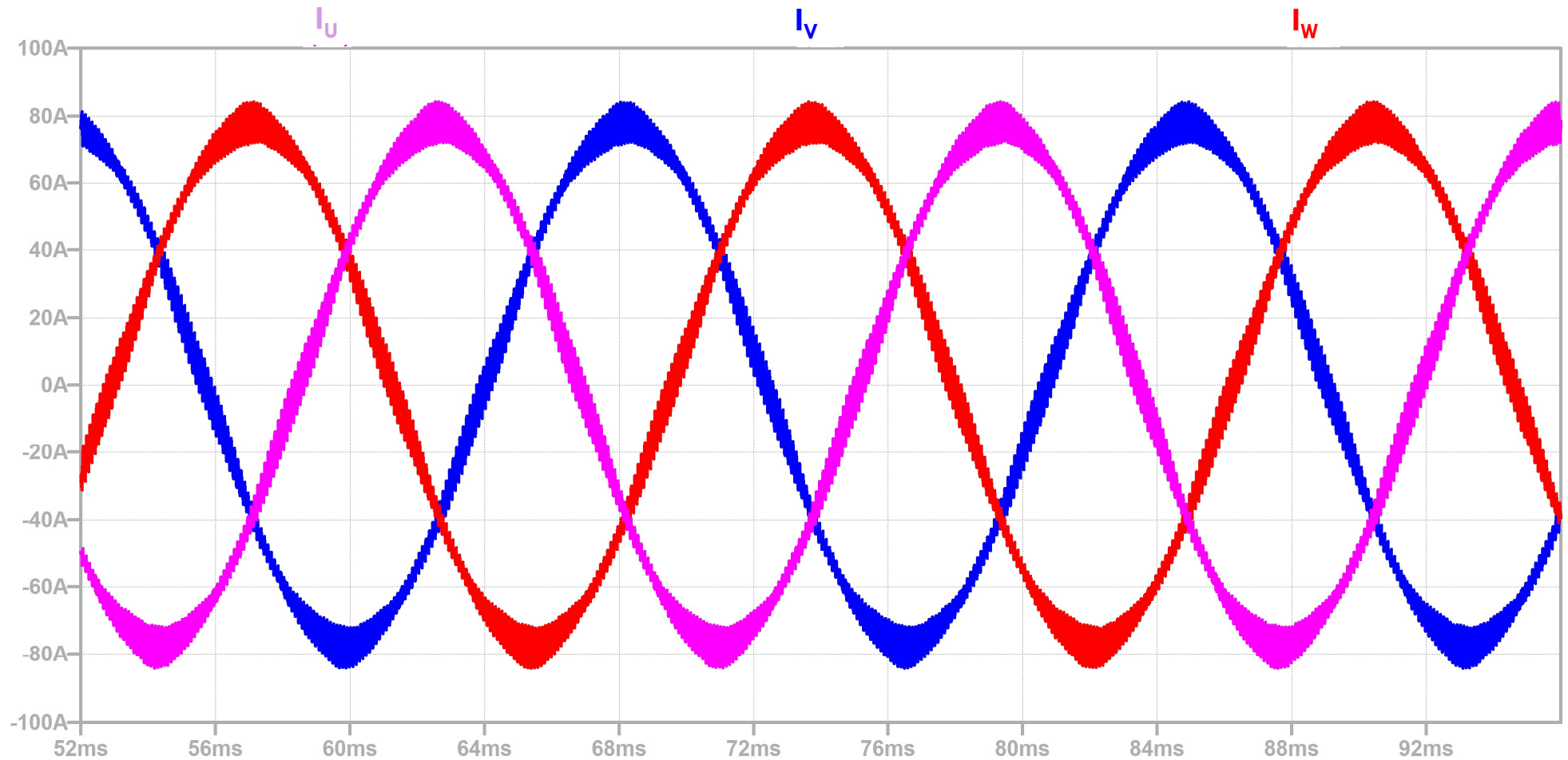
EXAMPLE

Phase (Line-to-Ground) Voltages between 118.8 ms and 119.5 ms



EXAMPLE

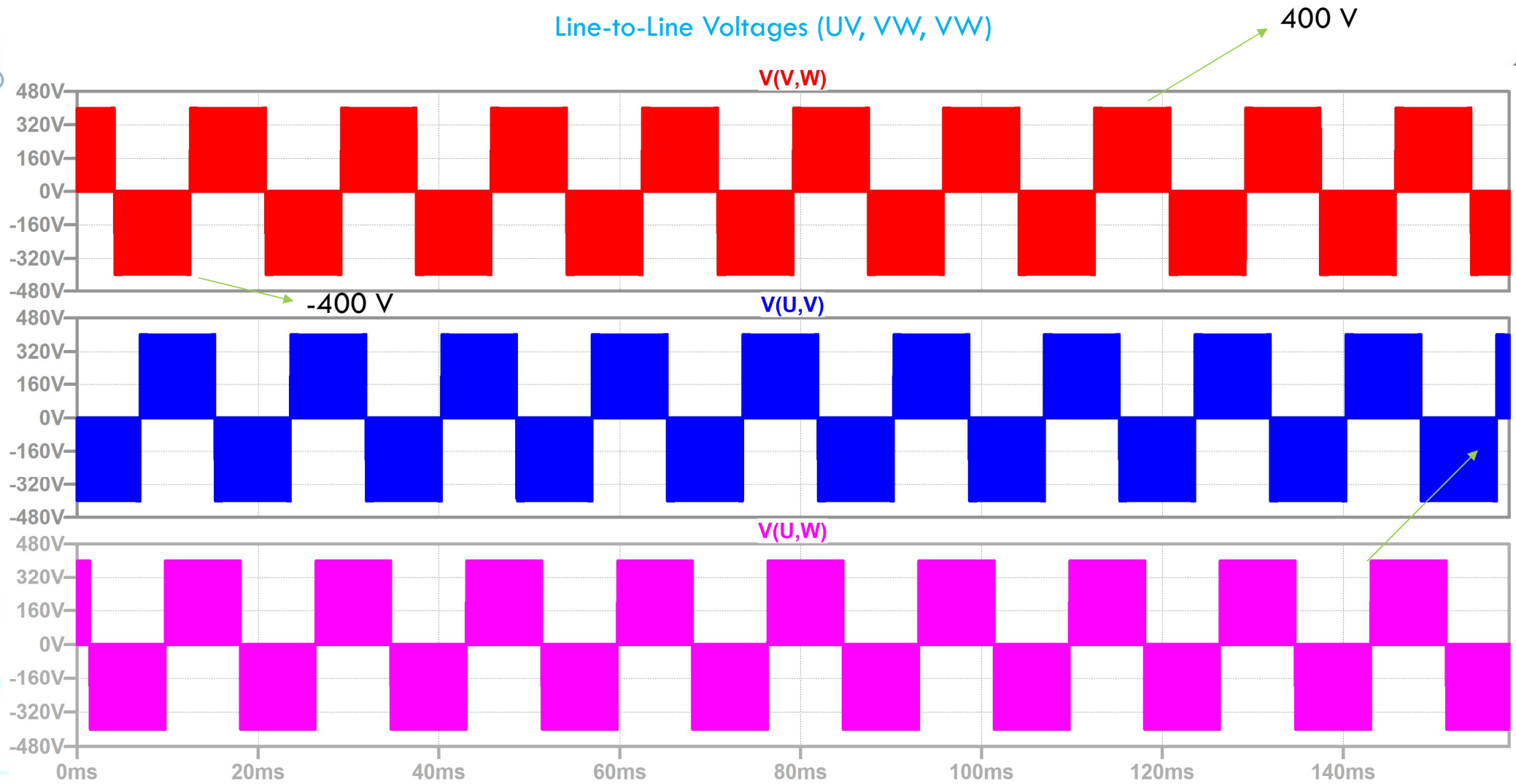
Phase Currents



Example: If the phase current U is positive, then MOSFET 1 injects current into the electric motor. If the phase current U is negative, then MOSFET 2 draws current from the electric motor.

EXAMPLE

Line-to-Line Voltages (UV, VW, VU)

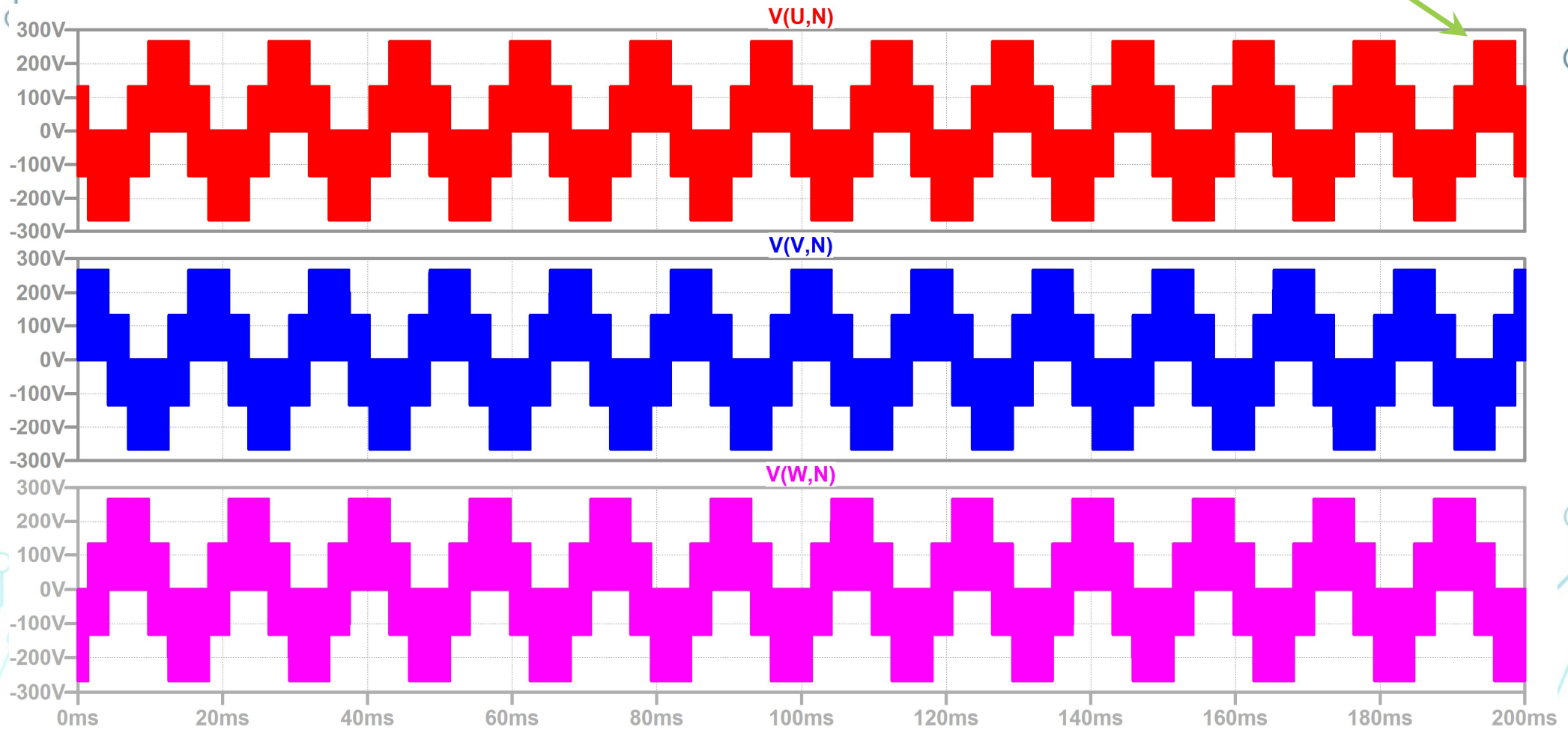


EXAMPLE

Line-to-Neutral Voltages (UN, VN, WN)

(not RMS!!!)

$$V_{LN,max} = (2/3) \cdot V_{DC} \approx 266 \text{ V}$$

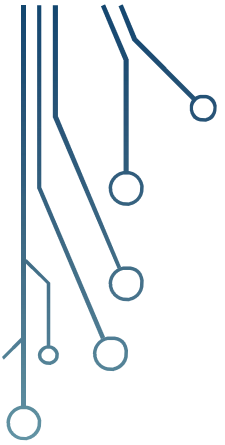


EXAMPLE

The phase voltage changes almost instantaneously with the switching frequency because it is directly imposed by the power electronic switches, such as PWM-controlled inverter devices. Each switching event modifies the applied voltage waveform within microseconds. The phase current, however, responds much more slowly because it is constrained by the inductance of the motor or load.

Inductance resists rapid changes in current, so high-frequency voltage components are largely filtered out. As a result, while the voltage can reflect every switching action, the current typically evolves over milliseconds and mainly follows the averaged or fundamental component of the voltage rather than the individual switching transitions.

$$V_L = L \frac{di_L(t)}{dt} \Rightarrow i_L = \frac{1}{L} \int V_L(t) \cdot dt$$



References

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